Ex No: Content Beyond Syllabus - 2: PRODUCER CONSUMER PROBLEM USING QUEUE

PROGRAM

```
#include <stdio.h>
#include <stdlib.h>

#define MAX 5

void producer();
void consumer();
void display();

int q[MAX], rear = 1, front = 1;

int main() {
    int choice;
    do {
        printf("\n1.Producer \n2.Consumer \n3.Display \n4.Exit");
        printf("\nEnter your choice: ");
        scanf("%d", &choice);
        switch(choice) {
            case 1:
                producer();
                break;
            case 2:
                consumer();
                break;
            case 3:
                display();
                break;
            case 4:
                exit(0);
            default:
                printf("\nEnter valid choice:");
        }
    } while(choice != 4);
    return 0;
}

void producer() {
    int value;
    if (rear > MAX) {
        printf("\nBuffer is full");
    } else {
        printf("\nEnter the piece of data : ");
        scanf("%d", &value);
        q[rear] = value;
        rear++;
    }
}

void consumer() {
```

```
    int x, i;
    if (rear == 1) {
        printf("\nBuffer is empty");
    } else {
        x = q[front];
        for (i = front; i < rear; i++)
            q[i] = q[i + 1];
        printf("\nThe consumed data is = %d", x);
        rear--;
    }
}

void display() {
    int count;
    if (rear == 1) {
        printf("\nThere are no data");
    } else {
        printf("\nThe data are");
        for (count = front; count < rear; count++) {
            printf("\n%d", q[count]);
        }
    }
}
```